

## Session 2: Valuation

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# Contact

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# Asset Valuation

- This morning
  - ▶ Assets (stocks, bonds, bank loans, etc.)
  - ▶ What they are used for (financing, risk sharing, etc.)
  - ▶ How they are created and traded (IPOs, exchanges, etc.)
- Now
  - ▶ What determines the value of assets, i.e., the price at which they trade
  - ▶ Ex.: Lemonade did an IPO at \$29 per share. How did investors come up with this valuation?

# Asset Valuation

- Value of asset = value of benefits generated by the asset
- “Benefits”
  - ▶ Usually in cash: dividends, interest, rents, royalties, etc.
  - ▶ Sometimes non-monetary: control rights, environmental impact
  - ▶ Called **cash flow (CF)** for brevity
- Two steps of valuation
  1. Forecast CF  $\leftarrow$  Data
  2. Value forecast CF  $\leftarrow$  Finance
    - ▶ CF are in the **future** and **uncertain**, so valuation must be adjusted accordingly

# Road Map

Valuing Risk-Free Assets

Valuing Risky Assets

Diversification

# Time Discounting

**Q1.** Do you prefer receiving €1 now or €1 in one year?



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**Q2.** How much do you value today €1 received in one year?

# Time Discounting

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- Time preference: future cash flows are discounted

Q2. How much do you value today €1 received in one year?

- This is called the **present value (PV)** of €1 in one year
- $r$  such that  $PV = 1/(1 + r)$  is the **discount rate**

# PV of Risk-Free Cash Flow

- More generally

- ▶  $D$  euros in  $t$  years has present value  $\frac{D}{(1+r)^t}$

- ▶ A stream of risk-free cash flows  $\{D_t\}_{t \geq 1}$  has present value

$$PV = \sum_{t=1}^{\infty} \frac{D_t}{(1+r)^t}$$

- Remarks

- ▶ Discount rate  $r$  is expressed per year, hence the power  $t$
  - ▶ “Risk-free,” “safe,” “deterministic”  $\neq$  “risky,” “stochastic”

# Valuing Risk-Free Assets

**Ex. 1** Consider two assets with the following risk-free cash flows

|         | Year 1 | Year 2 | Year 3 |
|---------|--------|--------|--------|
| Asset A | 10     | 10     | 10     |
| Asset B | 0      | 0      | 31     |

The discount rate is 5% per year

**Q1.** Which asset is more valuable?



2 minutes

# Valuing Risk-Free Assets

**Ex. 2** A US government bond promises 100 USD in one year. The discount rate is 4% per year

**Q1.** What is the price of the bond?



1 minute

# Valuing Risk-Free Assets

**Ex. 2** A US government bond promises 100 USD in one year. The discount rate is 4% per year

**Q1.** What is the price of the bond?



1 minute

**Q2.** You buy the bond and hold it for one year. What is your rate of return?



1 minute

$$\text{Return} \equiv \frac{\text{what you get} - \text{what you paid}}{\text{what you paid}} = \frac{\text{what you get}}{\text{what you paid}} - 1$$

# Risk-Free Assets

- Typical risk-free assets: debt with no risk of borrower default



Do you know of any such borrowers?

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Do you know of any such borrowers?

- ▶ Government bonds issued by fiscally sound countries (e.g., Germany)
- ▶ Corporate bonds issued by AAA-rated companies (e.g., Microsoft)

# Risk-Free Rate

- The **risk-free interest rate** (often called the risk-free rate) is the return on a risk-free asset, and also the discount rate for a risk-free asset
- Properties of the risk-free rate
  1. Same for all risk-free assets with cash flows at the same dates in the same currency
  2. Depends on the currency
  3. Depends on the cash flow horizon

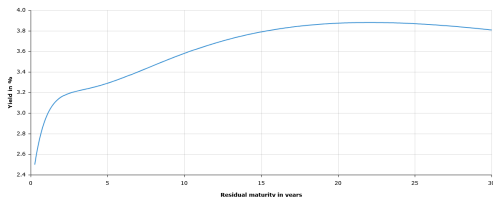
# Risk-Free Rate

2. Depends on the currency

| Country     | Latest yield |
|-------------|--------------|
| Australia   | 5.43%        |
| Austria     | 3.82%        |
| Belgium     | 4.38%        |
| Canada      | 4.00%        |
| Denmark     | 3.42%        |
| Finland     | 3.93%        |
| France      | 4.88%        |
| Germany     | 3.60%        |
| Greece      | 4.52%        |
| Ireland     | 3.67%        |
| Italy       | 4.66%        |
| Japan       | 3.10%        |
| Netherlands | 3.69%        |
| New Zealand | 5.13%        |
| Portugal    | 4.04%        |
| Spain       | 4.15%        |
| Sweden      | 3.29%        |
| Switzerland | 0.61%        |
| UK          | 5.46%        |
| US          | 5.31%        |

10-year rate. Source: [FT](#)

3. Depends on the cash flow horizon



Yield curve (a.k.a. term structure of interest rates) in euro. Source: [ECB](#)

## Remark: Market Efficiency

- “Value” and “price” are conceptually different
  - ▶ (Present) Value: how much investors should be willing to pay for an asset
  - ▶ Price: how much investors actually pay for an asset
- For now, we assume  $\text{price} = \text{value}$ , based on the following argument
  - If  $\text{price} < \text{value}$ , investors buy, pushing price up
  - If  $\text{price} > \text{value}$ , investors sell, pushing price down
  - ⇒ At market equilibrium,  $\text{price} = \text{value}$
- This property is called **market efficiency**. We will return to it on Thursday

# Road Map

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Valuing Risky Assets

Diversification

# Risk Aversion

**Q1.** Do you prefer receiving [€1 for sure] or [€0.5 or €1.5 with a fifty-fifty chance]?



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- **Risk aversion:** Investors value a risky cash flow less than a safe cash flow, holding expected cash flow fixed

⇒ Risky cash flows are discounted

# Valuing Risky Assets

- Most assets are **risky**: stocks, debt with default risk, etc.
- An asset with risky cash flow  $\tilde{D}$  in one year has present value

$$PV = \frac{E[\tilde{D}]}{1 + k}$$

where discount rate  $k = \underbrace{\text{risk-free rate}}_{\text{time discounting}} + \underbrace{\text{risk premium}}_{\text{risk discounting}}$

# Valuing Risky Assets

- More generally, an asset with risky cash flow  $\{\tilde{D}_t\}_{t \geq 1}$  has present value

$$PV = \sum_{t=1}^{\infty} \frac{E[\tilde{D}_t]}{(1+k)^t}$$

where:

discount rate ( $k$ ) = risk-free rate + risk premium

- Remarks
  - ▶ The PV formula may use a maturity-dependent discount rate:  $k_t$  instead of  $k$
  - ▶ The PV formula has other names: “Dividend Discount Model” in asset pricing, “Discounted Cash Flow” or DCF in corporate finance

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**Q.** Many companies do not pay dividends (e.g., Tesla). The PV formula imply their value is zero: true or false?

# Valuing Risky Assets Quiz

Consider two companies that will pay out dividends only once in their life (to keep the math simple), in one year from now:

- GreatTech's expected dividend is €400 million
- LameDuck's expected dividend is €100 million

Both companies have the same discount rate

**Q1.** Which company has the higher stock price?



1 minute

- a. GreatTech      b. LameDuck      c. Both have the same price

**Q2.** Which stock has the higher expected return?



1 minute

- a. GreatTech      b. LameDuck      c. Both have the same expected return

# Expected Return of Risky Assets

|                                 |
|---------------------------------|
| Expected return = Discount rate |
|---------------------------------|

- Easy to see with single payoff

$$\text{Realized return: } \tilde{r} = \frac{\tilde{D} - P}{P}$$

$$\text{Expected return: } E(\tilde{r}) = \frac{E(\tilde{D})}{P} - 1 = \frac{E(\tilde{D})}{\frac{E(\tilde{D})}{1+k}} - 1 = k$$

- The logic goes through with multiple payoffs

$$\text{Realized return: } \tilde{r}_{t+1} = \frac{\tilde{D}_{t+1} + (\tilde{P}_{t+1} - P_t)}{P_t}$$

We still have  $E_t(\tilde{r}_{t+1}) = k$  (proof on next slide)

## Proof that expected return = discount rate

Expected return from  $t$  to  $t+1$ :  $E_t(\tilde{r}_{t+1}) = \frac{E_t(\tilde{D}_{t+1}) + E_t(\tilde{P}_{t+1})}{P_t} - 1$

Using the PV formula:  $P_t = \sum_{\tau=1}^{\infty} \frac{E_t(D_{t+\tau})}{(1+k)^\tau}$

Expected price at  $t+1$ :  $E_t(\tilde{P}_{t+1}) = E_t \left[ \sum_{\tau=1}^{\infty} \frac{E_{t+1}(D_{t+1+\tau})}{(1+k)^\tau} \right]$

Using the law of iterated expectations:  $E_t(\tilde{P}_{t+1}) = \sum_{\tau=1}^{\infty} \frac{E_t(D_{t+1+\tau})}{(1+k)^\tau}$

Finally:  $E_t(\tilde{r}_{t+1}) = \frac{(1+k)P_t}{P_t} - 1 = k$

## Expected Return of Risky Assets

Discount rate = Risk premium + Risk-free rate = Expected return

- ⇒ Riskier assets have a higher risk premium
  - higher discount rate
  - higher expected return
  - lower price given expected future cash flows

# Historical Perspective

- Historical returns of main asset classes<sup>1</sup> (since 1960)

|                          | Return (% per year) |      |
|--------------------------|---------------------|------|
|                          | SD                  | Mean |
| Stocks                   | 17.3                | 11.3 |
| Corporate bonds          | 8.4                 | 7.8  |
| Real estate              | 19.3                | 12.3 |
| Commodities <sup>2</sup> | 24.9                | 8.7  |

⇒ Riskier assets have higher average returns



Commodities are puzzling: they have high volatility (i.e., SD) but not high average returns. Any idea why?

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<sup>1</sup>Doeswijk, Lam and Swinkels, “Historical Returns of the Market Portfolio,” *Review of Asset Pricing Studies*, 2019 [\[pdf\]](#)

<sup>2</sup>Commodities include mainly metals (e.g., gold), energy products (e.g., oil), and agricultural products

# Mini-Case: 2022 Stock Market Performance

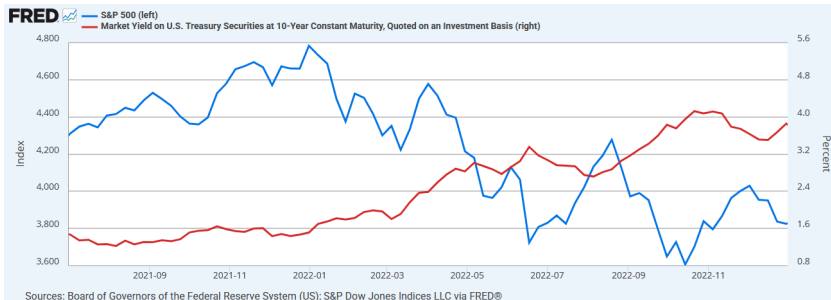
**Stocks fall to end Wall Street's worst year since 2008, S&P 500 finishes 2022 down nearly 20%**



What might explain it?

# Mini-Case: 2022 Stock Market Performance

**Stocks fall to end Wall Street's worst year since 2008, S&P 500 finishes 2022 down nearly 20%**



Higher (risk-free) interest rate  $\Rightarrow$  higher discount rate  $\Rightarrow$  lower stock value. **How much does it explain the stock price decline?**

## Mini-Case: 2022 Stock Market Performance

- Apply the PV formula to US stocks assuming a constant expected growth rate  $g$  of dividends:  $E[D_{t+\tau}] = (1+g)^\tau D_t$  for  $\tau \geq 1$

$$\implies PV_t = \sum_{\tau=1}^{\infty} \frac{D_{t+\tau}}{(1+k)^\tau} = \frac{(1+g)D_t}{k-g}$$

- Discount rate:  $k = \text{risk-free rate} + \text{risk premium}$
- The US 10-year rate rose from 1.5% at end-2021 to 3.5% at end-2022
- Calibrate risk premium  $\simeq 5\%$ , dividend growth rate  $g \simeq 4\%$

$\Rightarrow k - g$  increases from  $1.5 + 5 - 4 = 2.5\%$  to  $3.5 + 5 - 4 = 4.5\%$

$\Rightarrow$  This changes the PV by a factor of  $2.5/4.5 \simeq 0.56$ , i.e., reduces it by 44%



Directionally correct, but overstates the market decline. Why?

# Taking Stock

1. Price = expected cash flow discounted for time and risk
  2. Discount rate = expected return = risk-free rate + risk premium
- Remarks & next steps
    - ▶ 1. and 2. hold under the assumption of market efficiency (i.e., “the price is right”). We will challenge this assumption on Thursday
    - ▶ What determines the risk premium of an asset? We will answer this question tomorrow
    - ▶ To get there, we first need to examine *diversification*

# Road Map

Valuing Risk-Free Assets

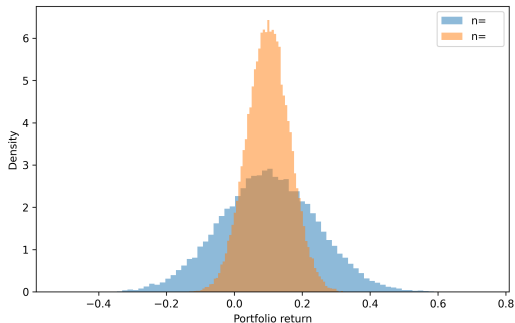
Valuing Risky Assets

**Diversification**

# Diversification

- A fund invests €100 million equally across stocks  $i = 1, 2, \dots, n$
- Each stock  $i$  has return  $r_i$  with  $E(r_i) = \mu$ ,  $V(r_i) = \sigma^2$ , and  $\text{Corr}(r_i, r_j) = \rho$  for all  $i \neq j$
- Portfolio return:  $r_P = \frac{1}{n} \sum_{i=1}^n r_i$

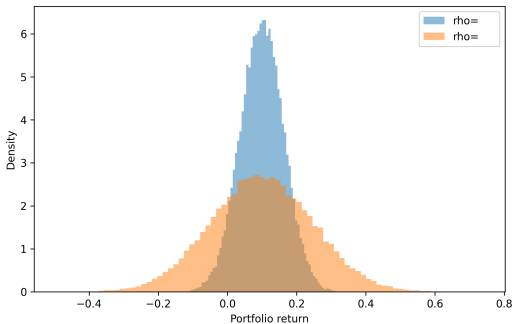
**Q1.** These histograms show the return distribution for  $n = 2$  and for  $n = 10$ . Which one is the orange distribution:  $n = 2$  or  $n = 10$ ?



$\mu = 10\%$ ,  $\sigma = 20\%$ ,  $\rho = 0$

# Diversification

**Q2.** These histograms show the return distribution for pairwise correlations  $\rho = 0$  and  $\rho = 0.5$ . Which one is the orange distribution:  $\rho = 0$  or  $\rho = 0.5$ ?



$n = 10, \mu = 10\%, \sigma = 20\%$

# Diversification

Therefore:

1. A portfolio holding a larger number of assets is better diversified, hence less risky (**Q. Is it always true?**)

# Diversification

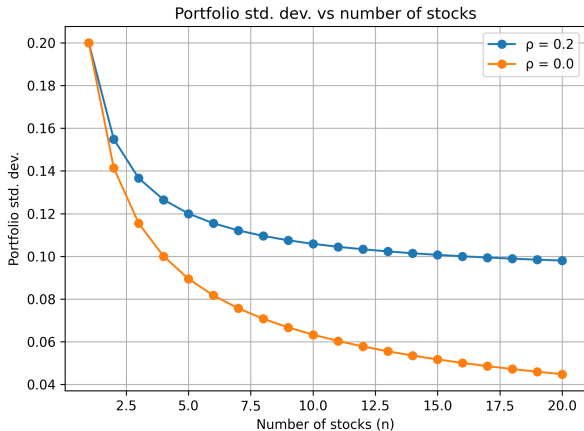
Therefore:

1. A portfolio holding a larger number of assets is better diversified, hence less risky (provided  $\rho < 1$ )
2. A portfolio holding assets with lower pairwise correlation is better diversified, hence less risky

**Q3.** Does portfolio risk go to zero when  $n$  goes to infinity?

- a. Yes      b. No      c. It depends

# Diversification



$\mu = 10\%$ ,  $\sigma = 20\%$

- When  $n \rightarrow \infty$ , portfolio risk depends on the **covariance** between assets held
- Proof: 
$$\text{Var}(r_P) = \left( \frac{1}{n} + \frac{n-1}{n} \rho \right) \sigma^2 \xrightarrow{n \rightarrow \infty} \rho \sigma^2$$

# Diversification

3. Risk uncorrelated across assets can be eliminated by diversification

Such risk is called **idiosyncratic (or diversifiable) risk**

4. Risk correlated across assets cannot be eliminated by diversification

Such risk is called **systematic (or non-diversifiable) risk**

# Diversified Portfolio Allocation

**Q.** What portfolio allocation would you advise for a long-term investor such as a pension fund or a sovereign wealth fund?

# Diversified Portfolio Allocation

Q. What portfolio allocation would you advise for a long-term investor such as a pension fund or a sovereign wealth fund?

⇒ Hold a well diversified portfolio. The maximally diversified portfolio is the **market portfolio**, which is invested in all investable assets

# Norwegian Sovereign Wealth Fund's Portfolio



# Until Tomorrow

- Practice problems on Slack

- See you tomorrow morning 🙄